

Archipelago (Island Chain / Distributed Warfare & Naval Control)

Executive Summary

This document outlines the distributed battle for distributed warfare, naval control, amphibious maneuver, and autonomous multi-node coordination. The terrain features scattered islands, coral reefs, deep-water channels, extensive offshore fortifications, future villages, and offshore naval platforms. The main emphasis: (Distributed Warfare) autonomous, distributed node task, naval air coordination, sensor fusion across water and land, and battle under fragmented terrain constraints. In AR (Real Life/AR), the main anchors at "4-5 m" relative distance, (Distributed Warfare) in physical area is approximately "1 km" of island-chain terrain. The features include multi-island objective nodes, naval headquarters, amphibious landing zones, and distributed control of sea lanes.

Design Goals & Gameplay

Distributed Multi-Island Warfare

Amphibious operations involve a series of multiple islands, each with unique terrain, resources, and defensive positions. Victory depends on coordinated multi-front pressure.

Naval Control & Sea-Lane Dominance

Deep-water channels, coral reefs, and offshore islands create naval chokepoints. Controlling sea lanes determines reinforcement speed and amphibious landing viability.

Amphibious Operations & Island Hopping

Players conduct coordinated island-hopping, using amphibious armor, drones, and infantry. Beaches, mangroves, and reefs create layered defensive zones.

Air Operations & Reconnaissance

VRX aircraft, drones, and the range of air platforms operate above the island chain. Airspace coordinates between cliffs and offshore platforms define movement and targeting.

Autonomy & Distributed Node Coordination

Autonomous units coordinate across multiple islands, selecting optimal nodes to attack or defend based on sensor data, enemy movement, and objective priority.

Sensor & Information Warfare

Sea spray, humidity, thermal distortion, sensor weather, and magnetic interference degrade sensors. Enemy towers on cliffs restore clarity. Destroying towers creates blind spots across the archipelago.

Logistics & Resources

Fuel caches, island depots, and offshore platforms serve as supply nodes. Convoys must navigate beach approaches and avoid ambush prone mangrove swamps.

Deployment & Progress

A head-on sensor range of 1000m enables players to measure distance, air superiority, or distributed amphibious mobility. Blue deploys from the southeast offshore staging platform, Red deploys from the northeast off-shore command island.

Spatial Constraints (Nreal AR)

Device Capabilities

View: 1080p, 90° FOV, 120-180 Hz per eye resolution. Optimal viewing distance ~4-5 m.

Playable Area & Scaling

100-150 m physical area ~ 1 km² island-chain zone (1:100 scale). Designed for tabletop AR.

Tracking & Occlusion

Unity uses Tracking for 3D horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view, assume minimal real-world occlusion.

Layout Overview

Coordinate System

Origin (0,0) = southeast corner.

Key Locations

Blue HQ (Offshore Staging Platform) — (0,0). 100-150 m Naval base, drone swarms, amphibious armor stacks.

Red HQ (Offshore Command Island) — (100,100). 100-150 m Naval base, drone swarms, amphibious armor stacks.

Central Island Cluster — (50,50). 100-150 m Three mid-sized islands with beaches, cliffs, and valleys.

Island Depot & Village — (10,10). 100-150 m Small bay, docks, interior floating nodes.

Offshore Sensor Tower — (100,100). 100-150 m High tower, sensor range, offshore armor stacks.

Deepwater Channel — (0,100). 100-150 m Primary naval chokepoint. Sealed defenses and coral hazards.

Offshore Fuel Base — (10,10). 100-150 m Low water sensor station with fuel depot.

Offshore Base A — (10,10). 100-150 m Low water sensor station with fuel depot.

Offshore Base B — (10,10). 100-150 m Low water sensor station with fuel depot.

Neutral Forward Outpost — (100,100) Drone resupply zone.

Chokepoints & Sightlines

Deepwater Channel

Major naval chokepoint, controlling it determines sea movement.

Mangrove Swamp Network

Interior foraging routes, mostly but slow.

Central Island Cluster

Multi-node control zone. Ideal for distributed warfare.

Offshore Fuel

Predictable but safe, vulnerable to overwatch from cliffs.

Sightlines

Offshore A provides long-range sensor coverage (100-150 m). Sea spray and humidity create fog-of-war pockets.

Deployment Zones

Blue Deployment: 100-150 m Offshore Staging Platform.

Red Deployment: 100-150 m Offshore Command Island, 100-150 m.

Neutral Forward Zone: Outpost at (100,100) for drone drops.

Assets & Environment

Terrain

Beaches, cliffs, coral reefs, mangroves, offshore platforms, island villages.

Vegetation

Palm trees, mangroves, shrubs, beach grass.

Buildings & Props

Military, oil rigs, radio dishes, sensor arrays, bunkers, industrial docks, generators, pipelines, Logistics: Docks, cargo crates, fuel drums, drones.

Effects

Sea spray, fog, humidity haze, wave impacts, thermal distortion.

Audio

Weapons, ship engines, sensor ping, radio chatter.

LOD & Budget

Asset List

Asset Name Count Volume Total Size

Environment 1 100 100

Buildings 15 15 15

Props 15 15 15

Textures 15 15 15

Audio 15 15 15

Total 2.2 MB

Texture Budget: 1000 for environment, 1000 for props, 1000 for audio, 1000 for compression. Target: 1000 MB.

Unreal / Nreal Integration

Engine & Plugins

Unreal Engine 5 + Nreal SDK (Optional). AR/VR/3DCore enabled.

Rendering

Forward rendering, stationary lights, baked GI, aggressive culling.

Assets

Single-point anchor for vector map.

Interaction

Phone gesture for projection, AR/VR as screen space widgets.

Multiplayer

Cloud-based architecture, offload AR/VR to server.

Implementation Plan & Testing

Milestones

1. Prototype Layout — Month 1 Build-out islands, reefs, mangroves, all anchor test.

2. Core Systems — Month 2 Build-out sea + land, amphibious autonomous, distributed node task.

3. AR Integration — Month 3 Build-out AR/VR, sensor fusion, AR/VR integration.

4. Content Fill — Month 4 Build-out terrain, textures, lighting, audio.

5. Gameplay & Audio — Month 5 Build-out 3D/AR/VR, sensor fusion, AR/VR integration, fog, thermal distortion.

6. Testing & QA — Month 6 Build-out AR/VR, sensor fusion, AR/VR integration.

Testing Plan

Autonomy

Verify real-world amphibious landings, distributed node coordination.

Comms/Sensors

Test relay distraction, fog-of-war updates, sensor interference.

Performance

100 Hz AR render on target device.

QA Checklist

Navigation, amphibious logic, distributed objective behavior, anchor drift, sensor accuracy.